Installation and Configuration Guide

Tessellate Gateway 3.4 for Linux servers · October 2026

1 Overview

Tessellate Gateway is a reverse proxy that routes API requests from client applications to internal services. It terminates TLS, applies rate limits per client key and writes a structured access log for every request. This guide describes a single-node installation; clustered deployments are covered in a separate document.

2 System requirements

Component	Minimum	Recommended
Operating system	64-bit Linux, kernel 5.4	Kernel 6.1 or later
Processor	2 cores	4 cores or more
Memory	2 GB	8 GB
Disk space	500 MB plus logs	20 GB on a separate volume
Network	One interface	Separate interfaces for clients and services

3 Installing on Linux

3.1 Prerequisites

- A user account with sudo rights on the target server
- Outbound HTTPS access to the download server, or the archive copied by hand
- A TLS certificate and private key for the public host name

3.2 Installation steps

1. Create a dedicated system user and group for the service.
2. Download the release archive and verify its checksum.
   - The checksum file is published next to each archive.
   - Stop if the verification reports a mismatch.
3. Unpack the archive under /opt and create a version-independent link.
4. Copy the example configuration file and adapt it as described in section 4.

The following commands perform steps 1 to 3 on a typical distribution:

```
sudo groupadd --system tessellate
sudo useradd --system --gid tessellate --home-dir /var/lib/tessellate tessellate
curl -fsSLO https://dl.tessellate.example/gw/3.4.2/tgw-3.4.2-linux-amd64.tar.gz
sha256sum -c tgw-3.4.2.sha256
sudo tar -xzf tgw-3.4.2-linux-amd64.tar.gz -C /opt
sudo ln -s /opt/tgw-3.4.2 /opt/tessellate
```

Note: Do not run the gateway as the root user. The service needs no elevated privileges once the listening port is above 1024 or the capability to bind low ports has been granted.

4 Configuration

The gateway reads a single YAML file at start-up. Changes to the upstream list can be applied without a restart by sending the reload signal, whereas changes to the listening address always require a restart.

```
# /etc/tessellate/gateway.yaml
server:
  listen: 0.0.0.0:8443
  workers: 4
  tls:
    certificate: /etc/tessellate/tls/gateway.crt
    private_key: /etc/tessellate/tls/gateway.key
upstreams:
  - name: billing
    url: http://10.20.0.15:9000
    timeout_ms: 2500
  - name: inventory
    url: http://10.20.0.22:9100
    timeout_ms: 4000
logging:
  level: info
  format: json
```

4.1 Main parameters

Key	Type	Default	Description
`server.listen`	address	0.0.0.0:8443	Address and port on which client connections are accepted.
`server.workers`	integer	number of cores	Worker threads handling requests; values above twice the number of cores rarely help.
`upstreams[].timeout_ms`	integer	3000	Time allowed for an upstream service to send the first byte of its response.
`logging.level`	string	info	One of debug, info, warn or error.
`logging.format`	string	text	Use json when the log is collected by an aggregation system.

4.2 Running as a service

Install the following unit file as /etc/systemd/system/tessellate.service, then enable and start the service:

```
[Unit]
Description=Tessellate Gateway
After=network-online.target

[Service]
User=tessellate
ExecStart=/opt/tessellate/bin/tgw --config /etc/tessellate/gateway.yaml
Restart=on-failure
LimitNOFILE=65536

[Install]
WantedBy=multi-user.target
```

5 Troubleshooting

Code	Meaning	What to do
TG-101	Configuration file not found	Check the path given with the --config option.
TG-204	Certificate and key do not match	Regenerate the key pair or correct the file paths.
TG-310	Upstream unreachable	Verify the address and the firewall rules between the gateway and the service.